



AWS Academy Data Engineering
Analyzing and Visualizing Data
Version 1.0.6

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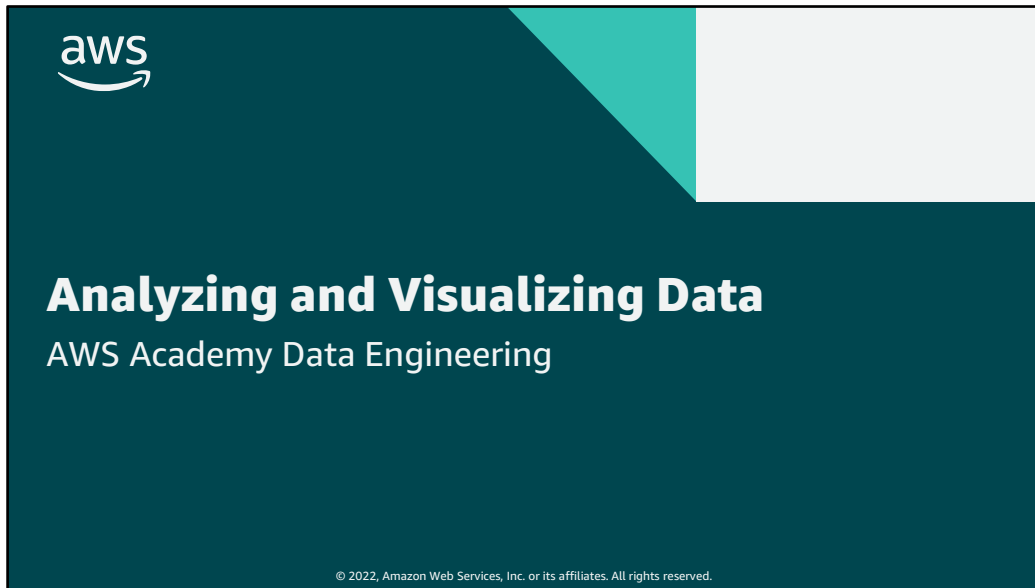
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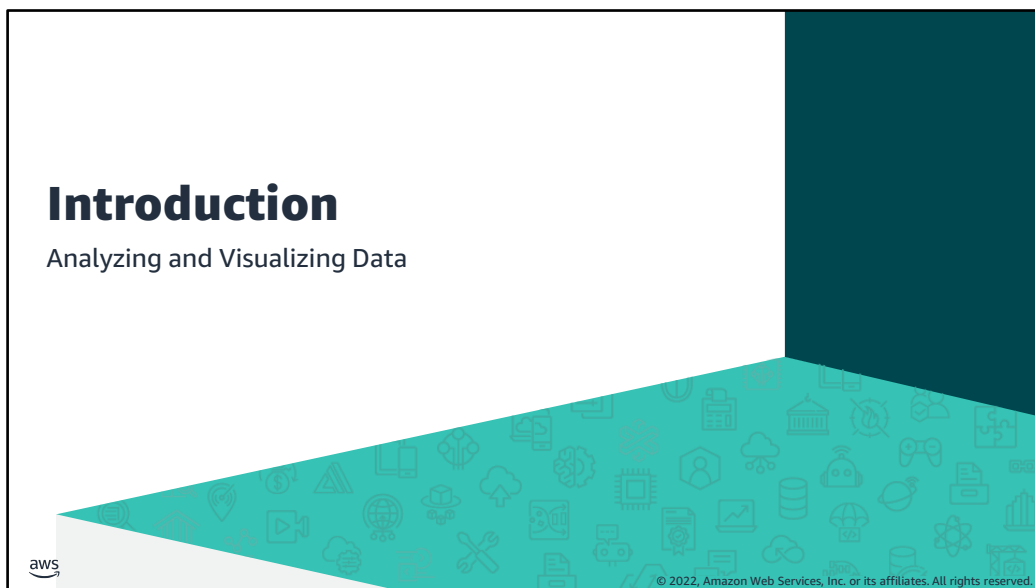
Contents

[Analyzing and Visualizing Data](#)

4



Welcome to the Analyzing and Visualizing Data module. This module focuses on AWS solutions to analyze and visualize data and factors to consider when you select a solution.



This introduction section describes the content of this module.

Module objectives



This module prepares you to do the following:

- List factors to consider when selecting analysis and visualization tools.
- Compare available AWS tools and services for data analysis and visualization.
- Determine the appropriate AWS tools and services to analyze and visualize data based on influencing factors (business needs, data characteristics, and access to data).

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3

In this module, the focus is on understanding what to consider when you select tools to analyze and visualize your data. The module compares three AWS tools and services for this purpose. You will also review a use case to examine the influencing factors and the selected AWS solution.

Module overview

Presentation sections

- Considering factors that influence tool selection
- Comparing AWS tools and services
- Selecting tools for a gaming analytics use case

Lab

- Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

Knowledge checks

- Online knowledge check
- Sample exam question



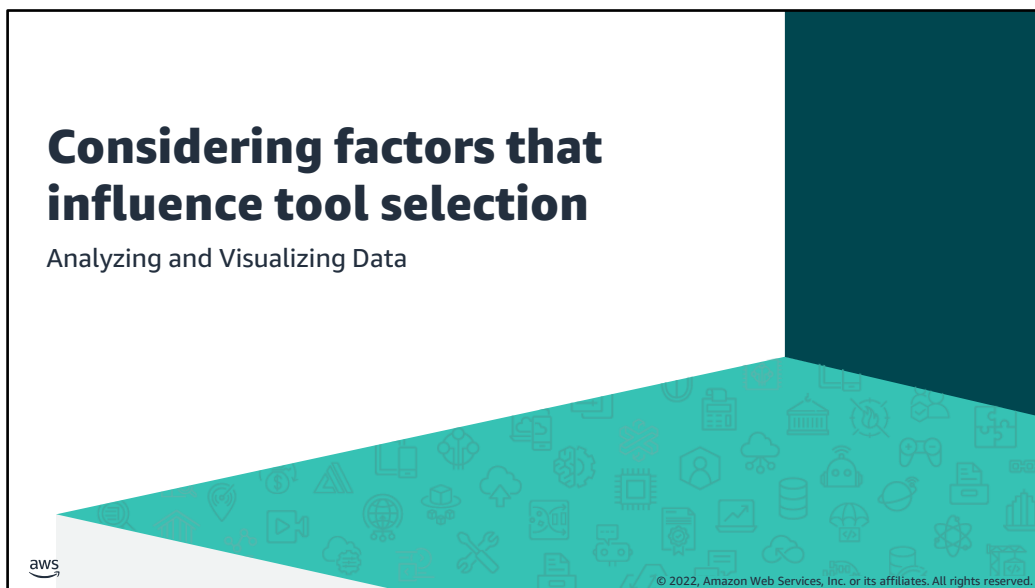
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4

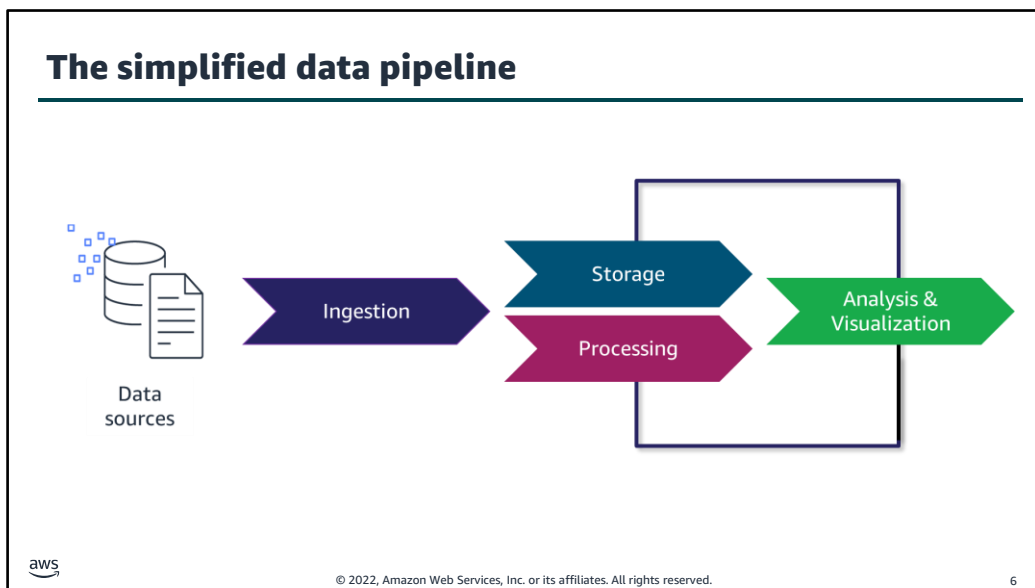
The objectives of this module are presented across multiple sections.

You will also complete a hands-on lab that uses Amazon Data Firehose, Amazon OpenSearch Service, and OpenSearch Dashboards to analyze and visualize data.

The module wraps up with a sample exam question and an online knowledge check that covers the presented material.



This section explains the factors to consider when you select analysis and visualization tools.



For accessibility: Data sources are ingested and move to storage and processing, followed by analysis and visualization. Processing between the storage, processing, and analysis and visualization phases can be iterative. **End of accessibility description.**

Throughout this course, you learned about the layers of a data pipeline. Raw data makes its way through the stages of the pipeline: ingestion, storage, processing, and analysis and visualization to produce insights about the data.

This simplified diagram illustrates where analysis and visualization is in the data pipeline. Although each layer is distinct, how data is analyzed and visualized is tied to how it is stored and processed.

Factors to consider when selecting tools

		
Business needs	Data characteristics	Access to data
Fully understand the business needs to be able to determine which data analyses and visualizations are needed to help develop insights.	Understand the type and quality of the data, and how often it is updated and processed.	Consider the data pipeline and who needs access to analyze and visualize data.

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Data engineers focus on the infrastructure that the data passes through. This three-part framework will guide you through the questions that you need to ask and the factors that you should consider when selecting data analysis and visualization tools and services. To build the correct pipeline, it's necessary to fully understand the desired outcomes based on business needs, the characteristics of the data, and who needs access to analyze and visualize data.

Business needs

- Which analyses are needed to help develop insights?
- What insights can be pulled from the data?
- What visualization would illustrate the insights?
- Does the consumer need to generate a report or interact with a dashboard?



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8

Keep the end goal in mind as you design the analysis and visualization layer of the data pipeline.

Business needs: Granularity of insight



Industry	Detailed Level	Aggregate Level
Finance	A finance manager wants information (such as revenue, costs, and profit margins) about their line of business.	A CFO wants similar metrics at an aggregate level across all lines of businesses. They want to be able to drill down to any line of business.
Marketing	A marketing manager wants to know about the number of leads, opportunities, and closed deals within an area (such as a postal code or city).	A CMO is interested in related metrics but at a broader level (such as a state or region).
Sales	A sales manager is focused on their sales pipeline and wants to know how long it takes to close an opportunity. They want to assess how many opportunities are needed to achieve quota targets.	A VP of sales wants similar information at an aggregate level. They want to be able to drill down to a sales representative or sales territory.



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9

The granularity of insight relates to the level of data that one has access to. A finance manager has access to data for their line of business, whereas the chief financial officer (CFO) has access to all the data.

Consider the grain size of the data that needs to be analyzed and visualized. Is it at a detailed level or an aggregate level?

For example, a marketing manager might want to know how various demand channels are performing. A chief marketing officer (CMO) might be interested in similar metrics at an aggregate level across all lines of businesses, with the ability to drill down into any line of business. The table shows examples of the granularity of insight for finance, marketing, and sales.

Business needs: Visualizing insights



KPIs	Relationships	Comparisons	Distributions	Compositions
Show performance in a particular area or function.	Establish or prove whether a relationship exists between two or more variables.	Show or examine how different variables change over time, or provide a static snapshot of how different variables compare.	Show how your data is distributed over certain intervals (based on clustering or grouping of data).	Highlight the various elements that make up your data—its composition.

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A key aspect of getting insight from your data is finding patterns. Patterns are often much easier to see in a graph or chart rather than staring at data in a table. The right visualization will help you gain a deeper understanding in a much quicker timeframe. A visualization might be produced as part of a report, or it might be used as part of an interactive dashboard where a user can drill down on something interesting.

Visualizations highlight one of the following five types of insights:

- Key Performance Indicators (KPIs) which are usually a single variable that measures how well you are doing. For example how many sales leads become sales.
- Relationships between two variables, for example, whether or not sales revenue is tied to marketing spending.
- Comparisons of how different variables change over time, for example showing month over month sales and web traffic.
- Distributions of data over certain clusters or groups, for example grouping customers by the number of purchases they've made.
- Compositions of elements that make up your data, for example, sales by region.

For more information, see the Data Visualization section of the aws.amazon.com website. A direct link is in the Content Resources section of the course.

Data characteristics

- How much data is there?
- At what speed and volume does it arrive?
- How frequently is it updated?
- How quickly is it processed?
- What type of data is it?



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11

Consider the volume, velocity, variety, veracity, and value of the data.

Data characteristics: Examples of data types

Volume and velocity

- **Historical analysis:**
Visualize a year's worth of sales data. Users can drill down by region and salesperson.
- **Streaming Internet of Things (IoT) data:**
Visualize the real-time error rates of sensors in a factory.

Variety and veracity

- **Structured data:** A relational database is queried to report on customer service tickets that were submitted in a specific period.
- **Unstructured data:**
Sentiment analysis is performed on customer service emails.

Value

- A business analyst uses **periodic reports** to showcase and report results to leadership.
- A DevOps engineer uses **self-service dashboards** to monitor and analyze performance in real time.



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12

Using the five Vs as an overarching framework, consider what types of data you are dealing with: historical analysis or streaming Internet of Things (IoT) data? Structured, semistructured, or unstructured data? How will the data be visualized—in a report or interactive dashboard?

Consider volume and velocity together as you make infrastructure decisions about how to collect, store, and process data. Consider how much data you need to ingest and how quickly you will ingest it.

Variety and veracity both relate to the data itself—what type of data is it and how trustworthy is it? The way that the data has been formatted and stored might affect how you analyze it.

Value is about ensuring that the outputs from analysis and visualization have business value, and knowing who needs to interact with and present the data visualizations.

Data characteristics: Two fraud detection use cases		
Data characteristic considerations	Use case 1: Rule-Based (Batch Pipeline)	Use case 2: ML in Real Time (Streaming Pipeline)
How much data is there?	Millions of transactions (kilobytes to terabytes)	Millions of transactions (bytes to megabytes)
At what speed and volume is it arriving?	In predefined intervals (minutes to multiple days)	In real time (milliseconds to seconds)
How quickly is it processed?	Minutes to hours	Milliseconds to seconds
What type of data is it?	Structured and semistructured	Unstructured and semistructured data
What value do insights from the data provide?	Historical reporting of fraud cases Reactive approach	Ability to detect fraud in real time Proactive approach

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For a fraud detection use case, consider the business needs first. For this purpose, two use cases are shown as examples here: rule-based fraud and machine learning (ML) in real time.

Next, consider the characteristics of the data by asking questions about the data. **How much data is there?** **What value do insights from the data provide?**

Rule-based fraud detection takes a reactive approach and provides a historical reporting of fraud cases. Structured data is ingested, analyzed, and reported in predefined intervals.

In contrast, the ML fraud detection example takes a proactive approach. It provides the ability to detect fraud in real time as millions of transactions go through the streaming pipeline.

Access to data

- Where does the data come from?
- Will the data need to be combined from multiple sources?
- Who needs access to the data and at what level? Who can access the tools?



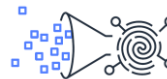
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14

Keep these questions in mind as you select analysis and visualization tools.

Access to data: Consider authorization level based on role

- A user's authorization to access data depends on their role in the organization.
- A business analyst or manager might be authorized to read the output that data engineers or data analysts create but not delete or update it.
- Follow the principle of least privilege. Give users the least amount of access and responsibility that are needed to complete their duties.

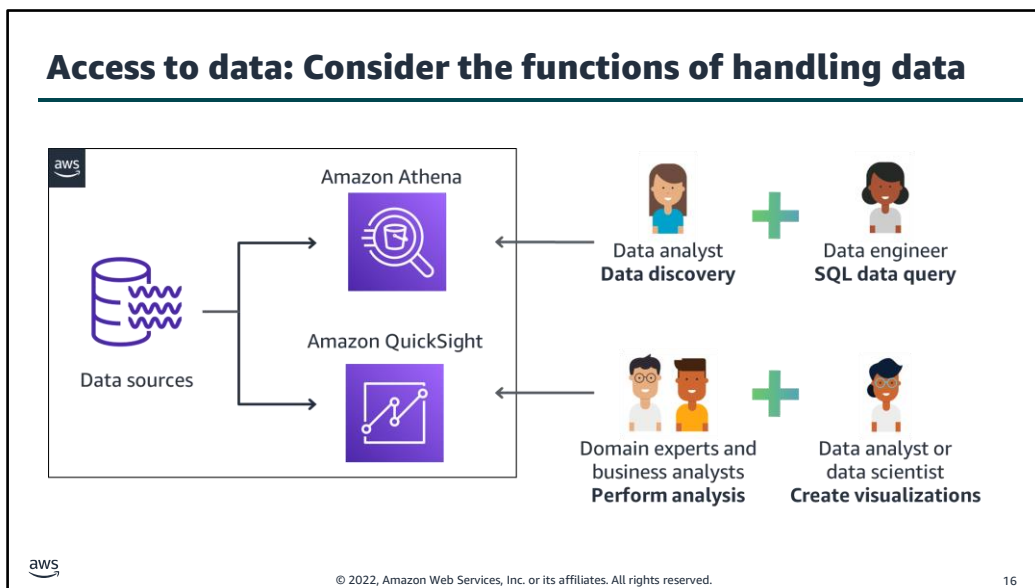


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15

Based on the principle of least privilege, a user's authorization to access data depends on their role in the organization. Consider this as you think about selecting tools and services to analyze and visualize data.

The Securing and Scaling the Data Pipeline module and the Data Analytics Lens of the AWS Well-Architected Framework have more details about securing your data analytics workloads.



A team might not include roles with the specific titles shown on this slide, but someone on the team is likely performing the same functions. A data analyst could work with a data engineer to query data and be able to perform interactive analysis by using Amazon Athena. Domain experts and business analysts could ask questions and explore the data with a data analyst or data scientist to create a visualization by using QuickSight.

**Key takeaways:
Considering
factors that
influence tool
selection**



- When you select analysis and visualization tools, consider the business needs, data characteristics, and access to data.
- Consider the granularity and format of the insights based on business needs.
- Consider the volume, velocity, variety, veracity, and value of your data.
- Consider the functions of individuals who will access, analyze, and visualize the data.

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17

Here are a few key points to summarize this section.

You learned about three factors to consider when you select tools to analyze and visualize data.

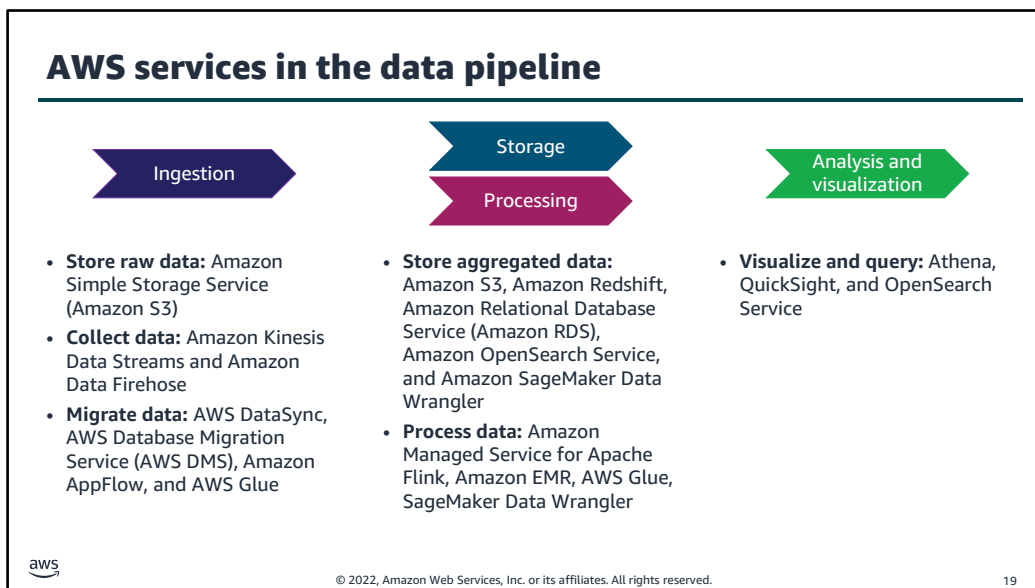
For business needs, consider the granularity and format of the insight.

For data characteristics, consider the volume, velocity, variety, veracity, and value of your data.

For access to data, consider the roles and functions of individuals who will access, analyze, and visualize the data.



This section compares AWS tools and services that are available to analyze and visualize data.



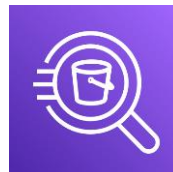
In this course, you have already encountered the tools and services that are related to the ingestion, storage, and processing layers of the data pipeline.

AWS tools and services that are commonly used to visualize and query data include Athena, QuickSight, and OpenSearch Service. In this section, you will learn about each service's features and compare them.

Amazon Athena

Amazon Athena is an interactive query service that provides the ability to use SQL to analyze data in Amazon S3. Athena includes the following features:

- Is serverless
- Provides the ability to combine data from multiple data sources
- Can be used for one-time queries
- Can be used from your favorite business intelligence (BI) tools (such as QuickSight)
- Can update data stored in Amazon S3 with Apache Iceberg integration



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20

Amazon Athena is an interactive query service in which you can use standard SQL to query and analyze data.

Athena is serverless. You don't need to set up anything, and you don't need to manage servers or data warehouses. You can use Athena to query data in place and combine data sources. This functionality is helpful for data analysts to perform one-time queries.

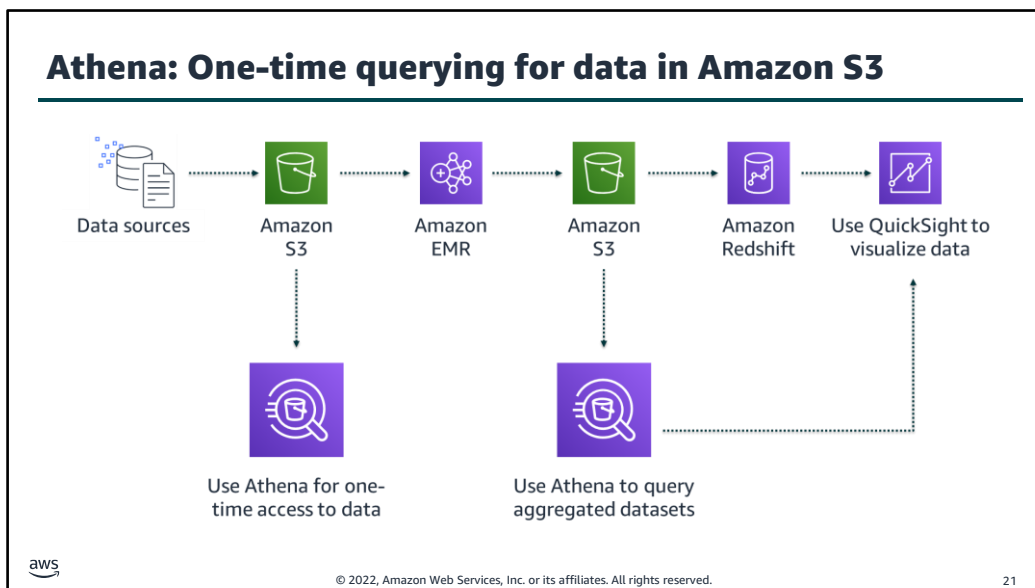


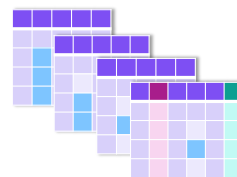
Image description: Data from multiple sources is put in Amazon S3, where Athena can be used for one-time queries. Amazon EMR aggregates the data and stores the aggregates in S3. Athena can be used to query the aggregated datasets. From S3, the data can be used in Amazon Redshift, where QuickSight can access the data to create visualizations. **End description.**

The diagram on the slide illustrates an example of using Athena to perform one-time queries of data that is stored in Amazon Simple Storage Service (Amazon S3).

You can upload data from multiple sources into Amazon S3 and query it using Athena for one-time access to data. You can also use Athena to query aggregated datasets.

Athena: Capability to update data in Amazon S3

- Users can use Athena to insert, update, and delete data that is stored in Amazon S3 (with the Apache Iceberg integration).
- Users can also track data versions automatically.
- The Apache Iceberg integration provides a way for continuous ingestion and updates.



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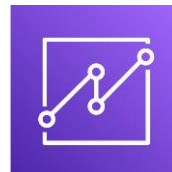
22

Using Athena, you can query data that is stored in Amazon S3, but you can't update the data.

Consider an example in which survey responses are stored in Amazon S3. Some responses aren't clear and need to be updated during the verification process. With the Apache Iceberg integration, the values are easily updated in Athena.

Amazon QuickSight

- Is a cloud-scale BI service to deliver easy-to-understand insights
- Connects to data in the cloud and combines data from many different sources
- Gives decision-makers the opportunity to explore and interpret information in an interactive visual environment
- Provides forecasting visualization capabilities
- Provides ability to ask questions using natural language with QuickSight Q feature



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23

QuickSight helps decision-makers interact with data in a visual environment. The service is designed to deliver easy-to-understand insights quickly.

For an introductory video about QuickSight and its interface, see the Amazon QuickSight Overview link in the Content Resources for the course.

QuickSight example use case



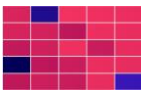
To visualize the sentiments, phrases, and tweets for a specific topic in QuickSight, you will view the following:



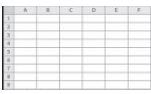
A donut chart with the overall sentiment for that specific topic sentiment



A word cloud of the phrases in the most dominant topic



A heat map of tweets that are associated with the dominant topics



A tabular view of related tweets



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24

In this case of using QuickSight, the situation is to analyze and visualize tweets that are related to a specific topic. The solution uses QuickSight to create a donut chart, word cloud, heat map, and table of the overall sentiment, most dominant phrases, and associated and related tweets.

For more information about this use case, see “Example Use Cases for Amazon QuickSight” in the Content Resources of the course.

QuickSight Q: Ask questions using natural language

When you need a visualization that isn't already in the dashboard, instead of waiting for the BI team to get back to you, use QuickSight Q to get immediate responses.

Example: You could say "Show me last year's weekly sales in California," and Q would provide an answer in a few seconds.

- Is a BI capability that's powered by ML
- Uses natural language processing
- Doesn't require you to build pre-defined data models or dashboards



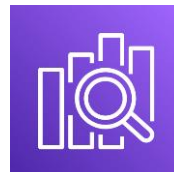
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25

Amazon QuickSight Q is powered by machine learning (ML) and provides a method for everyone in your organization to better understand your data. Users ask questions in natural language and receive accurate answers with relevant visualizations to help them gain insights from the data.

Amazon OpenSearch Service

- This managed service helps you deploy, operate, and scale OpenSearch clusters in the AWS Cloud.
- Use this open-source search and analytics engine for use cases such as the following:
 - Log analytics
 - Real-time application monitoring
 - Clickstream analytics



Amazon OpenSearch Service is integrated with visualization tools including OpenSearch Dashboards and Kibana.



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26

OpenSearch is an open-source search and analytics engine. Amazon OpenSearch Service is a managed service that provisions all the resources for your cluster and deploys it. The cluster is easy to set up, operate, and scale.

Example use case for OpenSearch Dashboards



Situation: Analyze and visualize support calls

Solution:

- Use Amazon S3, Amazon Transcribe, and Amazon Comprehend to get full transcripts of calls, keywords from the transcripts, and an overall sentiment of each call.
- Use OpenSearch Service and OpenSearch Dashboards to search and visualize the data:
 - A pie chart of the overall sentiment (positive, negative, neutral)
 - A bar chart of the keywords that are mentioned in the support calls
 - A histogram of when and how often the calls were made



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27

The example on this slide uses OpenSearch Dashboards to visualize data from support calls.

The business need is to be able to analyze and visualize support calls, such as what is the subject of each call? How many were positive? How many were negative? How can managers search or review the transcripts of these calls?

The solution uses other AWS services and tools to transcribe the support calls, and then uses OpenSearch Service and OpenSearch Dashboards to search and visualize the data.

For more information about this use case, see “Visualize with QuickSight Using Athena” in the Content Resources page of your course.

Comparison of AWS services for data analysis and visualization		
		
Athena	QuickSight	OpenSearch Service
• Interactive analysis using SQL • Analyze data directly • Start querying data instantly • Serverless	• Dashboards and visualizations • Build visualizations and dashboards for business analytics • Serverless	• Operational analytics • Search, explore, filter, aggregate, and visualize data in near real time • Fully managed service
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		28

Now that we have looked at the features of these AWS services, let's compare them. This comparison can help you know which service to choose for your particular use case.

Athena provides interactive analysis using SQL, while QuickSight is for dashboards and visualizations. OpenSearch Service is for operational analytics.

With Athena, you can start querying data instantly and directly in Amazon S3. With QuickSight, you can visualize data into insights quickly, and with OpenSearch Service, you can visualize data in near real time.

Athena and QuickSight are serverless, and OpenSearch Service is a fully managed service.

**Key takeaways:
Comparing AWS
tools and services**



aws

- AWS tools and services that are commonly used to query and visualize data include Athena, QuickSight, and OpenSearch Service.
- Athena is used for interactive analysis with SQL.
- Decision-makers can use QuickSight to interact with data visually and get insight quickly.
- OpenSearch Service is used for operational analytics to visualize data in near real time.

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29

Here are a few key points to summarize this section. You compared three AWS tools and services that are commonly used to query and visualize data.

Athena is used for interactive analysis with SQL. QuickSight is used to interact with data visually and get insight quickly. OpenSearch Service is used for operational analytics to visualize data in near real time.



In this section, you will apply what you learned about factors to consider when selecting tools, and the AWS tools and services that are available, to a gaming analytics use case.

Applying what you have learned in a use case for gaming analytics



The slide features three icons at the top. The first icon on the left shows a cloud with a magnifying glass over a line graph, with a database cylinder and a document below it. The middle icon consists of three purple squares: the top-left has a magnifying glass over a document, the top-right has a magnifying glass over a bar chart, and the bottom-center has a magnifying glass over a line graph. The third icon on the right shows a blue game controller above a console with a plus sign, a minus sign, and a circle button.

Factors that influence selection of analysis and visualization tools	AWS tools and services for analysis and visualization	Selecting tools for a gaming analytics use case
• Business needs • Data characteristics • Access to data	• Athena • QuickSight • OpenSearch Service	• Solutions based on a particular use case • Solutions based on personas in the use case

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Understand the use case and business needs to determine the potential solutions using AWS tools and services. Also, consider the personas and their functions within that context as you select AWS tools and services. See the next slides for an example.

Three personas in this use case for gaming analytics

How do different personas in a gaming company use AWS tools and services within different stages of the gaming analytics pipeline?



- **Analyst:** Explore and analyze player data
- **Business user:** Showcase and report results to leadership
- **DevOps engineer:** Monitor and analyze performance in real time

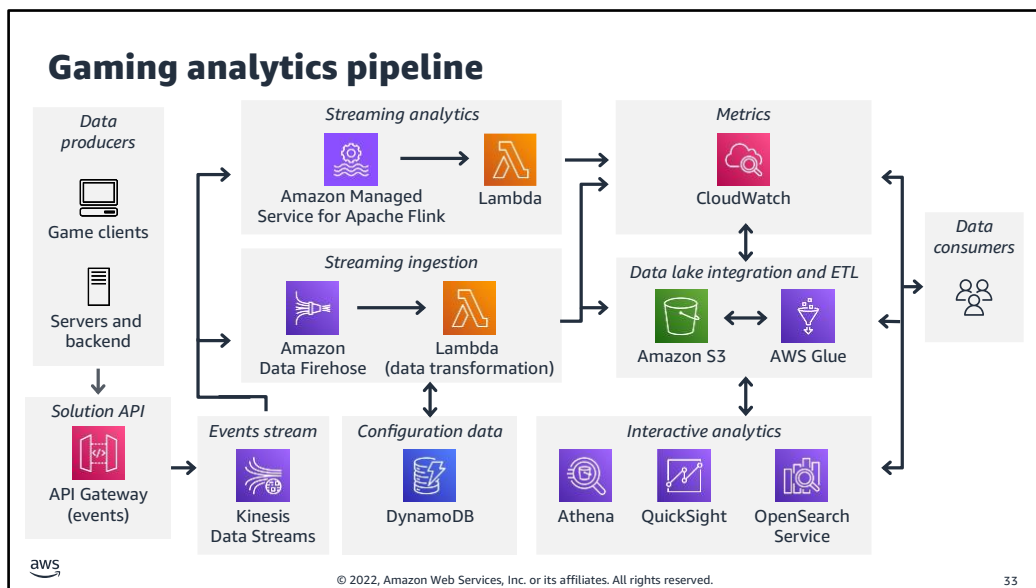


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32

A gaming company collects and analyzes data on an ongoing basis. The DevOps engineer monitors the health and performance of gaming servers, and the data analyst continuously analyzes player statistics and explores usage patterns. Business users also analyze data to showcase and visualize important information to leadership periodically.

Let's look at how three different roles—analyst, business user, DevOps engineer—might use a gaming analytics pipeline within the company.

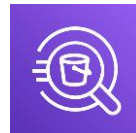


The servers and backend infrastructure that support the games produce data that Amazon Kinesis Data Streams ingests through Amazon API Gateway. Amazon DynamoDB acts as the store for the API keys. Amazon Managed Service for Apache Flink and Amazon Data Firehose process the data from Kinesis Data Streams in real time by invoking AWS Lambda functions. These functions process and transform the data. Amazon S3 then ingests the data from Amazon Data Firehose for further batch analysis.

Amazon CloudWatch provides real-time monitoring of the gaming servers, and the logs and metrics are stored in Amazon S3 for further analysis and visualization. Athena queries the log files that are stored in S3 or connects directly to CloudWatch to query the log data with SQL for analysis. QuickSight is used to create in-depth visualizations of server and backend performance metrics. These capabilities enable predictive analysis of the infrastructure burden and help identify areas of potential concern. Finally, engineers use OpenSearch Service to perform real-time additional monitoring as well as debugging and other troubleshooting.

Using Athena to explore and analyze player data

- **Business need:** Generate insights by querying daily aggregates of player usage data
- **Data characteristics:** Batch data for financial and geographical usage patterns
- **Data accessed:** Player purchase history, play history, and geographic information




Data
analysts



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34

The analyst in this scenario is interested in helping developers improve gameplay experience by analyzing player statistics, such as identifying which types of weapons and ammunition the players use. Similar methods can be used to visualize data about the geographic locations of the game's player base. Doing so would help developers determine regions that might need additional infrastructure to support a growing gaming population. This information could also help to identify possible cost savings in regions where the base isn't as large as initially predicted.

Using QuickSight to showcase and report results

- **Business need:** Visualize KPIs, such as average revenue per user, average revenue per paying user, retention rate, and conversion rate and forecasting
- **Data characteristics:** Combined data from multiple sources and aggregated to a high granularity level
- **Data accessed:** Player purchase history, play history, and geographic information



Business
users



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35

The business user's focus in this scenario is on summarizing, visualizing, and presenting metrics that are related to management concerns. The business user needs the capability to tailor reports to management's desired scope, which makes QuickSight an excellent tool for data visualization.

KPIs for gaming companies include metrics such as conversion rate. The conversion rate measures the number of users who have made a purchase during a specific time period. The ability to turn free users into paying users is a high priority for investors. QuickSight can help to showcase such metrics, including forecasting, effectively.

Using OpenSearch Service to monitor and analyze performance in real time

- **Business need:** Monitor health and performance, and analyze performance for predictive load balancing
- **Data characteristics:** Large volumes of streaming telemetry data and server logs, including structured and unstructured data
- **Data accessed:** Access logs and performance data for game servers




DevOps
engineers



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36

The DevOps engineers perform real-time monitoring of the gaming servers and backend infrastructure to ensure that gameplay is smooth and uninterrupted. They use OpenSearch Service to visualize the health of gaming servers and backend performance metrics in real time. These capabilities enable predictive analysis of the infrastructure burden and help identify areas of potential concern.

**Key takeaways:
Selecting tools for
a gaming analytics
use case**



- This use case showcased the granularity of visualized insights:
 - Daily batch aggregates of client usage patterns (Athena)
 - Consolidated aggregate KPIs for leadership (QuickSight)
 - Continuous health and performance monitoring (OpenSearch Service)
- Keep the influencing factors in mind when you select AWS tools and services. Multiple solutions exist to meet the business needs of data analysis and visualization.

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37

Here are a few key points to summarize this section.

The use case showcased the granularity of visualized insights. Daily batch aggregates of client usage patterns were visualized by using Athena. Consolidated aggregate KPIs were visualized for leadership through QuickSight. Continuous health and performance monitoring was visualized with OpenSearch Service.

Keep the influencing factors in mind when you select AWS tools and services. Multiple solutions exist to meet the business needs of data analysis and visualization.



You will now complete a lab. The next slide summarizes what you will do in the lab, and you will find the detailed instructions in the lab environment.

Lab introduction:
Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards



- In this lab, you will analyze user activity and access patterns for a university bookstore website by using streaming log data.
- You will build an index for CloudWatch log events in OpenSearch Service.
- Then, you will use OpenSearch Dashboards to visualize the data:
 - Create a pie chart to show the operating systems and browsers that visitors use to consume the website.
 - Create a heat map to show how users are referred to product pages.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform during this lab.

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39

Access the lab environment through your online course to get additional details and complete the lab.

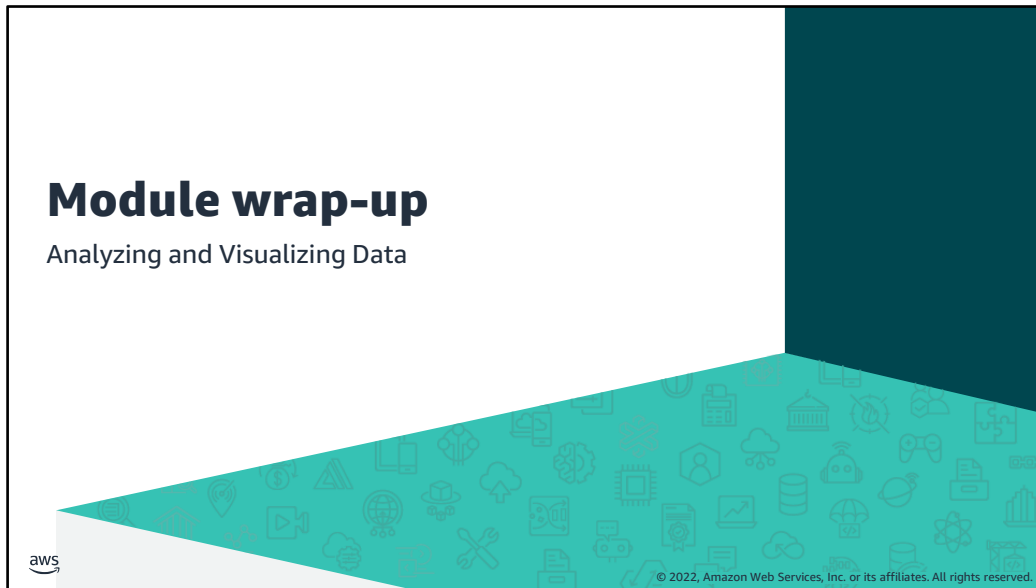
Debrief: Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

- In this lab, you had to log in to OpenSearch Service with credentials other than your AWS credentials.
 - Which service managed user authentication to OpenSearch Service?
 - Which service managed user authorization?
 - Why were AWS services needed as part of the architecture in this lab?
- What is the purpose of index patterns in OpenSearch?
- What was the purpose of the Lambda function named **os-demo-lambda-function**?
- What was the purpose of the **agent.json** file on the Apache web server?
- Why was CloudWatch needed as part of the overall solution?



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40



This section summarizes what you have learned and brings the Analyzing and Visualizing Data module to a close.

Module summary

This module prepared you to do the following:

- List factors to consider when selecting analysis and visualization tools.
- Compare available AWS tools and services for data analysis and visualization.
- Determine the appropriate AWS tools and services to analyze and visualize data based on influencing factors (business needs, data characteristics, and access to data).



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42

This module described factors that influence selecting analysis and visualization tools. You compared AWS tools and services for data analysis and visualization. Ultimately, the module has prepared you to determine appropriate AWS tools and services to analyze and visualize data based on influencing factors (business needs, data characteristics, and access to data).

Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

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43

Use your online course to access the knowledge check for this module.

Sample exam question

A company's web application produces 250 GB of clickstream data per day, which is stored in Amazon S3. The company wants to use the data to find out how quickly the application's webpages loaded during the last month. They want to be able to compare month-to-month data to gain insights. Which data analysis and visualization solution would provide these capabilities while also minimizing the cost and complexity?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- 250 GB of **clickstream data** per day
- **How quickly** the application's webpages loaded during the **last month**
- **Compare month-to-month data** to gain insights
- **Minimizing the cost and complexity**



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44

Select the most cost effective data analysis and visualization solution that you would consider as a data engineer.

Sample exam question: Response choices

A company's web application produces 250 GB of **clickstream data** per day, which is stored in Amazon S3. The company wants to use the data to find out **how quickly** the application's webpages loaded during the **last month**. They want to be able to **compare month-to-month** data to gain insights. Which data analysis and visualization solution would provide these capabilities while also **minimizing the cost and complexity**?

Choice	Response
A	Use Amazon OpenSearch Service to analyze and OpenSearch Dashboards to visualize the data.
B	Use Apache Spark to analyze the data and Amazon EMR Hive to visualize the data.
C	Use Amazon Managed Service for Apache Flink to analyze the data and Microsoft Excel to visualize the data.
D	Use Amazon Athena to analyze the data and Amazon QuickSight to visualize the data.



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45

Use the key words that you identified on the previous slide, and review each of the possible responses to determine which one best addresses the question.

Sample exam question: Answer

The correct answer is D.

Choice	Response
D	Use Amazon Athena to analyze the data and Amazon QuickSight to visualize the data.

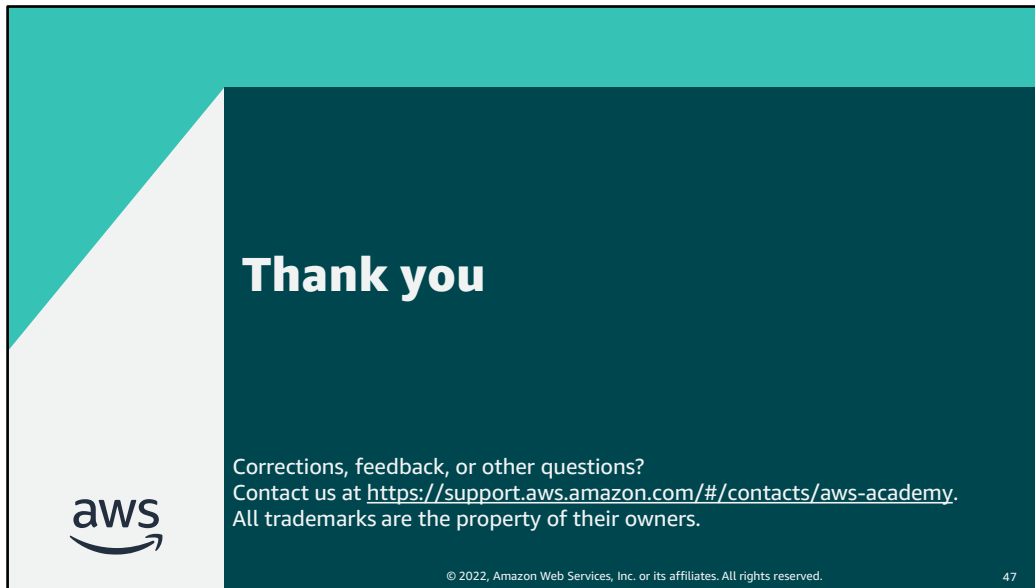
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Choice A (Use OpenSearch Service and OpenSearch Dashboards) is not the most cost-effective solution. It isn't cost-effective to collect and keep large amounts of historical data to use with OpenSearch Service and OpenSearch Dashboards.

Choice B (Use Apache Spark and Amazon EMR Hive) is not an appropriate solution for data analysis and visualization. Although Spark can be used to process the data, Amazon EMR Hive does not offer capabilities for data visualization.

Choice C (Use Amazon Managed Service for Apache Flink and Microsoft Excel) does not meet the requirement to minimize complexity. Needing to download the data increases complexity and prevents the solution from being scalable.

Choice D (Use Athena and QuickSight) is the most cost-effective and appropriate solution for the analysis and visualization of historical, clickstream data over time.



That concludes this module. The Content Resources page of your course includes links to additional resources that are related to this module. Thank you for completing this module.